

Informal Regulation and Housing Supply: Evidence from Chicago’s Aldermanic Privilege

Jacob Herbstman[†]

September 14, 2026

Abstract

Land-use regulation in cities depends not only on the formal rules, but also on the individual politicians who administer them. Many U.S. cities allow substantial informal discretion in the development approval process, often exercised by the local politician representing the district where development is proposed. I study informal regulatory discretion in Chicago, where “aldermanic privilege” gives city council members substantial influence over development in their wards. I construct alderman-level “stringency scores” from permit processing times and use complementary event-study and boundary-discontinuity designs to estimate the effects of these informal regulations on housing supply and prices. Assignment to a more stringent alderman lowers permitting activity by 12%, while multifamily density is 13–17% lower immediately on the more-stringent side of a ward boundary. I also find that home prices are 3.0% higher on the more-stringent side, while rents are 1.8% higher but not statistically distinguishable from zero. These results show that informal regulation can distort housing supply and potentially affect citywide housing affordability, even in areas where the formal zoning rules are similar.

JEL Codes: R14, R31, R52

Keywords: Housing supply, land-use regulation, local political economy

[†]Corresponding author. Harris School of Public Policy, University of Chicago, 1307 E. 60th Street, Chicago, IL 60637, United States. Email: jherbstman@uchicago.edu.

I thank Eric Zwick, Justin Marlowe, Joshua Gottlieb, Asad Khan, Chris Berry, Ayo George, and seminar participants at the Harris School PhD workshop, Economics Department Student Applied Micro Lunch, and the Notre Dame Housing Economics Conference for helpful comments and suggestions.

1 Introduction

Housing costs in the United States have risen sharply, and a large literature points to land-use regulation as a central cause. Restrictive zoning, permitting frictions, and inclusionary zoning requirements limit housing supply, raise prices, and distort the allocation of labor across cities (Glaeser, Gyourko and Saks, 2005; Saiz, 2010; Turner, Haughwout and Van Der Klaauw, 2014; Hsieh and Moretti, 2019; Duranton and Puga, 2023; Soltas, 2024; Soltas and Gruber, 2026). However, formal zoning rules are not the only determinant of what gets built. Informal regulations, often exercised through the discretionary power of local politicians, also play a significant role in shaping development patterns. Many major cities in the United States have a fragmented regulatory approval system that includes the discretionary judgment of the local city councilmember or a similar politician.¹ This potential for individual discretion to differentiate otherwise similar regulatory environments can induce uncertainty in the development process and potentially lower housing supply in cities (Mayer and Somerville, 2000*a*; Hilber and Robert-Nicoud, 2013; Khan, 2021; Mast, 2024). Informal discretion from individual politicians can substantially lengthen the development timeline. When developers face long and unpredictable timelines, projects shrink, marginal investments become unviable, and housing supply falls (Cunningham, 2006; Bloom, 2009; Oh, Yang and Yoon, 2024). Despite this evidence, there is little empirical work on how individual politicians use these informal regulatory powers to shape housing supply.

This paper studies this form of informal regulatory power in the context of Chicago’s “aldermanic privilege,” the unwritten but influential authority of each of Chicago’s 50 city council members (aldermen) to unilaterally block or approve development in that member’s ward. Chicago has no city charter, so there is no statutory basis for this power (McClintock, 2025). It operates through an unspoken mutual understanding: other aldermen vote with the local member on land use matters in exchange for similar support in their own

¹Analogous forms of council-member deference in land-use decisions have been documented in Philadelphia, Los Angeles, and New York City (Burnett and Kogan, 2014; Jacobson, 2023; Mitre-Becerril and MacDonald, 2024; New York City Charter Revision Commission, 2025).

wards, city agencies follow the local alderman’s lead, and developers understand that major projects require the alderman’s blessing (Thale, 2005; Einstein, Glick and Palmer, 2019; Khan, 2021). This creates sharp changes in regulatory authority at ward boundaries, even when the neighborhood and zoning are quite similar. Two parcels on opposite sides of the same street, in the same neighborhood, with the same zoning, can face very different regulatory regimes depending on which alderman represents them. I use this sudden change in regulatory authority to study how informal regulations affect housing supply and prices.

To measure the regulatory environment each alderman creates, I construct “regulatory stringency scores” from processing times for permits likely to involve substantial aldermanic discretion, which include new construction, renovations, and demolitions. I deem these “high-discretion” permits since they are the most likely to require aldermanic approval. The motivation for using processing time variation is simple: delay makes projects less valuable, and a developer who expects years of meetings, redesigns, and uncertainty may build a smaller project or be deterred from building at all (Kydlund and Prescott, 1982; Mayer and Somerville, 2000*a*; Paciorek, 2013; Gyourko, Hartley and Krimmel, 2021; Soltas and Gruber, 2026).

I first use the 2015 ward redistricting to test whether permitting changes when a census block is reassigned to a different ward. This design holds location fixed while changing the assigned alderman. Using this within-census-block variation, I find that reassignment toward greater stringency reduces the annual number of high-discretion permits recorded by the city by about 12% during 2015–2020. This result is driven asymmetrically by movement toward more stringent aldermen: these census blocks see a 20.7% reduction in high-discretion permits, compared with a 7.0% increase for blocks reassigned to a more lenient alderman. I find no statistically significant effect on low-discretion permits, consistent with an effect concentrated in permits more exposed to aldermanic discretion.

I then examine whether new housing is built at lower density under more stringent aldermen. Following Bayer, Ferreira and McMillan (2007) and Kulka, Sood and Chiumenti

(2026), I employ nonparametric regression-discontinuity designs at ward boundaries to estimate effects on new construction density. I find a 9% reduction in FAR and a 14% reduction in DUPAC on the more-stringent side of ward boundaries, with slightly larger effects for multifamily new construction.

I also compare rents and sale prices across ward boundaries. I find a sharp increase in home prices across the boundary of about 3.0%. I also find a 1.8% increase in listed rents, but the estimate is imprecise and not statistically significant. Although these results are more suggestive, their direction is consistent with lower supply on the more-stringent side and with informal regulation affecting affordability.

This paper contributes most directly to work on local political control and fragmented land-use authority. Fischel (2009) argues that homeowners use local institutions to restrict development and protect property values, and subsequent work shows how decentralized jurisdictions can adopt restrictions that are locally rational but collectively costly (Hilber and Robert-Nicoud, 2013; Ortalo-Magné and Prat, 2014; Monarrez and Schönholzer, 2023; Mast, 2024; Bordeu, 2025; Kulka, Sood and Chiumenti, 2026). I show that politicians within a fragmented system can create different regulatory environments even when formal rules are shared.

In Chicago specifically, Freemark (2020) studies formal upzoning and Khan (2021) finds that aldermen approve smaller rezonings near ward boundaries where the benefits to their constituents are lower. I add to this literature by showing that the effects of informal regulatory discretion extend beyond formal zoning or statutory upzonings and appear in permitting activity and the density of new construction, not only in zoned density.

The paper also contributes to work on permitting delays and time-to-build, since the effects of these informal regulations are often realized through development uncertainty and delays. Development lags are a first-order determinant of housing supply (Mayer and Somerville, 2000*a,b*; Paciorek, 2013). Recent work finds that rules-based planning shortens construction timelines and that developers place substantial value on lots with ready-to-issue

permits (Lyons and Sweeney, 2025; Soltas and Gruber, 2026). I use alderman-specific variation in permitting time and connect this to both permit activity and completed housing density.

2 Institutional Background and Data

2.1 Aldermanic Privilege and the Development Process

Chicago’s City Council consists of 50 aldermen, each representing a single ward. Chicago’s uniquely informal norms give each alderman broad authority over development within ward boundaries. This practice, known as “aldermanic privilege” or “aldermanic prerogative,” rests on custom rather than formal law, in part because Chicago has no city charter (McClintock, 2025). On matters affecting a single ward, other aldermen defer to the local representative, and city agencies follow the alderman’s lead (Thale, 2005; Zhang, 2011). The convention is strong enough that aldermen have often been described as “little mayors” governing their wards as personal fiefdoms (Einhorn, 1991). One alderman described the system candidly: “I often liken the City of Chicago to a feudal system, where the mayor is sort of a de facto king. And each alderman is the lord of their individual fiefdom.” (Keefe, 2013)

The formal mechanism. Two reforms narrowed this authority in 2019. A new construction code repealed a rule that had generally required the Department of Buildings to notify the local alderman of permit applications and wait ten days. Mayor Lightfoot’s Executive Order 2019-2 separately directed city departments not to defer to aldermen when municipal law did not require it. Neither reform eliminated aldermanic influence: other notification requirements remained, departments continued to solicit aldermanic input, and aldermen retained City Council authority over zoning and other legislative approvals (City of Chicago, 2019; Mayor’s Press Office, 2019; Schutz, 2019). Lightfoot captured the problem directly: “If you’ve got 50 separate fiefdoms that you have to deal with on issues like affordable housing,

or just getting a sign on your building, it makes doing business with the city of Chicago damn near impossible.” (Spielman, 2019b) The Metropolitan Planning Council made the same point from the developer side: “before a development even gets to the point of being indefinitely delayed, many never make it past ‘Go’ because of required aldermanic letters of support.” (Metropolitan Planning Council, 2019)

How prerogative shapes development. Aldermanic discretion enters well before formal permit issuance and continues after a project is filed.

Pre-application. Experienced Chicago developers know that projects requiring rezoning or smaller modifications need the alderman’s support before formal papers are filed. Although not legally required, aldermanic support is all but essential in practice: without it, a developer risks investing heavily in design, legal, and engineering work before facing rejection at the political stage. One anonymous Chicago architect described the dynamic: “To get the alderman, you have to get the neighborhood group, and that’s where the trouble comes in. It’s fraught with risk, even to have this conversation.” (Giangreco, 2024)

Rezoning. For projects requiring a change to the zoning code, the alderman’s role is effectively dispositive. Although zoning amendments formally require a City Council vote, the council “routinely shows unquestioning deference to local aldermen even in the absence of any articulated local concern” (Chase, 2023). Equally consequential is the alderman’s ability to preemptively *downzone* parcels to foreclose future development. Joseph Schwieterman, a DePaul University professor who studies Chicago land use, noted that the practice “creates real uncertainty for real estate developers when downzoning becomes ad hoc or driven by the desire of an alderman to settle scores.” (Gallun, 2019) Documented episodes include Ald. Brendan Reilly’s proposals to downzone a River North site where a developer had proposed a 60-story tower and a separate site to block a 615-room hotel, and a Pilsen downzoning orchestrated by Ald. Solis that stripped a parcel of its residential designation to kill a planned 465-unit apartment project.

Permit processing. For projects that proceed by right under existing zoning, the alderman retains substantial informal influence over the Department of Buildings permit queue. The corruption case against Ald. Ed Burke illustrated the mechanism in an extreme form: federal prosecutors documented how his staff withheld routine permits—including a driveway permit for a fast-food restaurant—until the property owner directed legal business to Burke’s law firm (Spielman, 2019a). The case made visible what developers had long understood: in Chicago, permit timing is another lever aldermen can use to delay and suppress projects they oppose. In interviews, a former alderman described substantial use of the permit process to deter developers from trying to overrule the alderman on rezoning and planning matters earlier in the process.

A recent example. Developer Fern Hill began community outreach in 2021, proposing a roughly 500-unit mixed-income tower at a transit-accessible site in the 2nd Ward, with approximately 100 affordable units and an estimated \$2.5 million in annual property tax revenue. After four years and more than ten community meetings, Ald. Brian Hopkins announced in January 2025 that he would not advance the project, citing opposition from residents of neighboring high-rises. He was explicit about the source of his authority: “Ultimately I am the decision maker on this. I’m not going to hide from that. This is a zoning decision and in Chicago we have something called aldermanic prerogative.” (Achong, 2025) Within two weeks the developer resubmitted a 349-unit proposal, cutting 151 units (30 percent) and 101 feet from the tower; Hopkins then approved it (TRD Staff, 2026). The episode captures both margins this paper studies: the approved project is smaller than what the developer proposed (intensive margin), and the four-year ordeal signals to other developers the cost of attempting large projects in wards with active aldermanic gatekeeping (extensive margin).

Federal scrutiny. In 2023, the U.S. Department of Housing and Urban Development (HUD) concluded that aldermanic prerogative operates as a “local veto” that allows wards

to block, deter, or downsize affordable housing, especially in majority-white areas (Chase, 2023); outside groups have made similar civil-rights arguments (Shriver Center on Poverty Law, 2019). The question of whether to reform or abolish aldermanic privilege remains a central issue in Chicago politics.²

2.2 Building Permits and the Permit Process

The City’s public data contain permits recorded by the Department of Buildings since 2006. Each record reports when the application began and when the permit was issued, but cannot capture projects abandoned before they progress far enough to generate a permit record. I use the universe of “high-discretion” permits from 2006–2022 that have valid geographic coordinates and nonnegative processing times. Online Appendix A.3 describes the separate sample used to estimate stringency scores in the event study. I divide permits into two groups based on how much aldermanic discretion they are likely to face.

High-discretion permits are permits for projects where aldermen are most likely to matter: new construction, renovation or alteration, demolition, porch construction, and reinstated permits.³ These are the permit types most likely to draw community opposition and face delays due to community meetings or direct aldermanic intervention. *Low-discretion permits* are all remaining permit types outside this high-discretion group, excluding signs. I use them as a placebo comparison group because they are generally less exposed to discretionary aldermanic intervention.

High-discretion permits take longer to process and vary more across aldermen: their mean processing time is 57.3 days, compared with 26.2 days for low-discretion permits (Figure 1). Processing time is negatively associated with ward-month permit volume for high-discretion permits but weakly positively associated for low-discretion permits, consistent with the former capturing regulatory friction rather than general administrative capacity.

²See <https://news.wttw.com/elections/voters-guide/2023/races/city-council>, where each city council candidate was asked specifically about their stance on aldermanic prerogative.

³Porch construction permits have received extra scrutiny since the 2003 balcony collapse.

Online Appendix Table C.3 reports the corresponding statistics.

2.3 Regulatory Stringency Scores

I construct the stringency score using permit records from 2006–2022. I first adjust processing times for observed differences across wards, locations, permits, and months, then estimate the remaining alderman-specific differences. The first stage regresses log processing time at the permit level on ward-level demographic controls (median household income, racial composition, homeownership rate, population), permit-location geographic controls (distance to the central business district (CBD), lakefront proximity, Chicago Transit Authority (CTA) rail access), lagged ward-month permit volume, and fixed effects for year-month, permit type, and review type:

$$\log(\text{Processing Time}_i) = X_i'\theta + \phi_{m(i)} + \psi_{c(i)} + \xi_{v(i)} + \varepsilon_i, \quad (1)$$

where i indexes permits, X_i contains the observed permit and location controls, and $m(i)$, $c(i)$, and $v(i)$ denote the permit’s year-month, permit class, and review type. Online Appendix Table B.1 reports the coefficients on the first-stage controls.

The second stage averages the adjusted processing times by ward and month, then regresses those averages on alderman indicators and the same lagged ward-month permit-volume control. Ward-months are weighted by the number of permits observed:

$$\bar{\varepsilon}_{wm} = \mu + \alpha_{a(w,m)} + \chi \text{Volume}_{w,m-1} + \nu_{wm}. \quad (2)$$

Here $a(w, m)$ denotes the alderman serving ward w in month m . The raw alderman fixed effects $\hat{\alpha}_a$ can be noisy for aldermen with few permits. I therefore apply empirical Bayes shrinkage (Kline, Rose and Walters, 2022, 2024) to pull imprecise estimates toward the grand mean, then standardize the scores, denoted S_a , to have mean zero and unit standard deviation.

Online Appendix Figure B.1 maps the stringency scores in 2014 and 2022, shows their distribution, and reports a low within-ward correlation between predecessor and successor aldermen ($\rho \approx 0.16$). The scores vary across wealthy and poor neighborhoods and across the North and South Sides, supporting an alderman-specific rather than ward-level interpretation.

2.4 New Construction Data

I construct a citywide dataset of new residential construction in Chicago between 2006 and 2022 using publicly available data from the Cook County Assessor’s Office. I combine residential improvement characteristics with commercial valuation records for properties assessed under multifamily model groups. Together, these files report dwelling units, stories, building area, and land area. I locate each project using Assessor parcel records, assign it to a ward using City of Chicago shapefiles, and define the construction date as June 15 of the recorded year built since I cannot observe an exact date. For the boundary-discontinuity analyses, I divide ward boundaries into approximately 1,320-foot segments and assign each project to the nearest segment. Because some ward boundaries are several miles long, the segments ensure that compared projects are near both the same boundary and each other. I keep projects with positive lot area, positive building area, and at least one dwelling unit. The FAR and DUPAC analyses use the same projects with both measures available.

Because zoning measured today may reflect subsequent changes, I assign each project the broad zoning group in force on its assigned construction date. Online Appendix A.4 describes the historical zoning reconstruction.

For each project I compute two standard density measures: Floor Area Ratio (FAR), defined as total building area divided by lot area, and Dwelling Units Per Acre (DUPAC), defined as dwelling units divided by lot acreage. I measure each project’s distance to its nearest ward boundary.

Table 1 presents summary statistics. The descriptive sample contains 13,635 new resi-

dential projects throughout Chicago with both density measures available, including 2,582 classified as multifamily. I report the multifamily sample separately because aldermanic discretion is likely to matter most for larger projects.

2.5 Rental Listings

RentHub provides scraped rental listings. I use Chicago listings from 2014 through December 2022 and collapse repeated scrapes to one floorplan-month observation, using the median asking rent observed that month. Rents are expressed in 2022 dollars using the monthly Chicago CPI-U.

2.6 Home Sales

Home sales come from Cook County Assessor records. I use filtered transactions of single-family homes, townhouses, and small multifamily buildings (2–6 units) from 2006 through 2022. Prices are expressed in 2022 dollars using the monthly Chicago CPI-U. Full 14-digit property index numbers (PINs) determine location and date-specific ward assignments, and same-year Assessor records provide property characteristics. I exclude transactions whose exact day is unavailable when an adjoining ward changes aldermen or the ward map changes during the reported month.

Online Appendix A provides additional detail on the rental and home sales data.

3 Conceptual Framework

3.1 Developer Problem

Delay makes projects less valuable. A developer who expects years of meetings, redesigns, and uncertainty may respond by building a smaller project, waiting, or walking away (Kydland and Prescott, 1982). The model below turns that intuition into a simple time-to-build

problem.

Consider a developer choosing the scale $q \geq 0$ (units of housing) of a project on a given site. Construction costs are $C(q) = F\mathbf{1}\{q > 0\} + cq + \frac{\eta}{2}q^2$ with $F, \eta > 0$, so building requires a fixed cost and marginal costs rise with project size. The developer can instead leave the site unbuilt and earn zero. Let P denote the local price level, taken as given at the project level.

Let S index the stringency of the local regulatory environment. Stringency maps into an *effective time-to-build* $T(S)$ with $T'(S) > 0$. The term $T(S)$ summarizes entitlement delay, redesign requirements, negotiation costs, financing frictions, and regulatory uncertainty that push expected revenue further into the future. If revenue arrives upon project completion, the developer's expected discounted revenue per unit is $e^{-rT(S)}P$, where r is the relevant discount or financing rate. The developer solves

$$\max_{q \geq 0} e^{-rT(S)}Pq - C(q).$$

Conditional on building, the first-order condition gives the interior scale

$$q^I(S, P) = \frac{e^{-rT(S)}P - c}{\eta}.$$

At this scale, the project's profit is $(e^{-rT(S)}P - c)^2 / (2\eta) - F$. The developer therefore builds only when the project covers its fixed cost, and the optimal choice is

$$q^*(S, P) = \begin{cases} \frac{e^{-rT(S)}P - c}{\eta}, & \text{if } e^{-rT(S)}P - c \geq \sqrt{2\eta F}, \\ 0, & \text{otherwise.} \end{cases}$$

For projects that are built, $\partial q^* / \partial T(S) < 0$. Longer or more burdensome regulatory processes therefore reduce project scale and can also push marginal projects from “build” to “do not build.”

3.2 From Developer Choices to Empirical Predictions

Indexing sites by i , total housing supply is $Q = \sum_i q_i^*(S_i, P_i)$. Higher $T(S_i)$ reduces Q through two direct margins: projects that are built at lower density and marginal projects that are no longer built. These are the main predictions I take to the data. The flow of permits requiring aldermanic review should fall when stringency increases (Section 4.1), and sites in more stringent regulatory environments should have lower completed density (Section 4.2).

With downward-sloping demand $P'(Q) < 0$, a supply contraction can also raise housing costs. That prediction is less sharp in the data because prices may adjust over a wider geography, but I find suggestive evidence that prices are higher on the more-stringent side of ward boundaries in Section 4.3.

3.3 Interpretation of the Stringency Score

In the model, these frictions enter through the index $rT(S)$. A higher value means revenue arrives later and is worth less today. That affects both the developer's intensive-margin density choice and the extensive-margin decision to build or not.

The aldermanic stringency score S_a constructed in Section 2.3 measures the alderman-specific component of observed processing time. I interpret a higher S_a as a more stringent regulatory environment and thus a higher effective time-to-build $T(S)$. Other dimensions of regulatory friction, such as impact fees, design mandates, or community benefit requirements, would enter as additional costs in $C(\cdot)$ and reinforce the same qualitative predictions. The core idea is that longer observed processing times for permits on larger discretionary projects are a noisy but informative signal of delay that lowers density, reduces aggregate supply, and may raise prices.

4 Results

4.1 Permit Activity After Redistricting

I first ask whether permit activity changes when the same location is assigned to a different alderman. Chicago’s 2015 ward redistricting moved some census blocks toward wards represented by more stringent aldermen and others toward wards represented by more lenient aldermen. Nearby blocks that remained in their original ward provide the comparison group. The design therefore holds the physical location fixed while changing its assigned regulatory environment. The primary analysis sample keeps boundaries for which both the origin- and destination-ward aldermen serving in 2014 remained in office when the new map took effect, since newly elected aldermen’s development preferences could be endogenous to prior development activity in the ward. I also require each block to have at least one high-discretion permit during the five years before redistricting. Without that restriction, any later permit on a block with no previous activity would mechanically appear as an increase.

The outcome is the number of high-discretion permits recorded on each census block in each year from 2010 through 2020. The outcome therefore measures when projects observed in the permit data first appeared in the process. These are the high-discretion permits defined in Section 2.2.

I estimate a Poisson pseudo-maximum likelihood event study that treats moves toward greater stringency and greater leniency as opposite changes:

$$\log \mathbb{E}[Y_{by} \mid \cdot] = \mu_b + \lambda_{g(b),y} + \sum_{t \neq -1} \beta_t Z_b \mathbf{1}\{y - 2015 = t\}. \quad (3)$$

Here Y_{by} is the permit count for block b in calendar year y , μ_b is a block fixed effect, and $\lambda_{g(b),y}$ is a ward-pair-by-year fixed effect for the ward pair $g(b)$ containing block b . Event time is $t = y - 2015$, with $t = -1$ omitted. Let ΔS_b be the destination-minus-origin difference in standardized aldermanic stringency for block b . For blocks that change wards,

$Z_b = \text{sign}(\Delta S_b)$; it equals zero for blocks that remain in the same ward. I classify moves using the 2014 incumbents in each block’s origin and destination wards, with scores estimated only from permits filed between 2006 and 2014. Using the 2014 incumbents separates ward reassignment from changes in representation associated with the 2015 elections. This specification imposes equal-and-opposite effects in logs for assignments toward more stringent and more lenient aldermen. Panel A of Figure 2 plots the coefficients β_t . The pre-trends are not distinguishable from zero. The pooled 2015–2020 estimate is -0.127 ($\text{SE} = 0.049$), equivalent to an 11.9% reduction in the annual number of high-discretion permits associated with a move toward greater stringency. Panels B and C relax the symmetry restriction by estimating the two directions separately. The decline in permitting activity comes mostly from moves toward more stringent aldermen, which reduce high-discretion permits by 20.7%; blocks assigned toward more lenient aldermen experience a 7.0% increase. Online Appendix Table C.2 reports the corresponding pooled estimates without the symmetry restriction and shows the results are nearly identical.

This asymmetry suggests that aldermanic discretion operates through uniquely stringent aldermen blocking activity that would otherwise occur, rather than uniquely lenient aldermen allowing activity that formal city regulations would otherwise block. This distinction matters for the likely aggregate effect of informal regulatory discretion on housing supply because it is possible *ex ante* that a few particularly pro-development aldermen approve more housing than a centralized system would. The results instead suggest that informal aldermanic discretion reduces housing supply relative to a more centralized counterfactual.

4.2 Density at Ward Boundaries

I next examine the individual new construction projects described in Section 2.4 on opposite sides of the same ward boundary, moving the unit of analysis from census blocks to projects. Projects within 500 ft of the same boundary segment face similar neighborhood amenities and demand conditions, but different aldermen. Following Bayer, Ferreira and McMillan

(2007) and Kulka, Sood and Chiumenti (2026), I divide signed distance to the boundary into 100 ft bands and compare projects in each band with projects in the nearest band on the less-stringent side. I estimate the following specification:

$$\begin{aligned} \log(Y_{isy}) = & \sum_{k \in \mathcal{K}, k \neq [-100, 0)} \kappa_k \mathbf{1}\{d_{is} \in k\} + X_i' \theta \\ & + \zeta_{z(i,y)} + \delta_s + \gamma_y + \varepsilon_{isy}. \end{aligned} \tag{4}$$

Here Y_{isy} is FAR or DUPAC for project i , assigned to boundary segment s and built in calendar year y . The running variable d_{is} is negative on the less-stringent side and positive on the more-stringent side. The set $\mathcal{K}$ contains ten 100 ft bands from -500 to 500 ft. The band from -100 to 0 ft is omitted, so $\kappa_{[0,100)}$ compares projects immediately across the boundary. The vector X_i contains the White population share, the Black population share, median household income, the bachelor's-degree share, and the homeownership rate on the project's side of the boundary. The model also includes the zoning group in force when the project was built, boundary-segment fixed effects, and construction-year fixed effects. Standard errors are clustered by ward pair.

Figure 3 plots the distance-band estimates. The omitted band is normalized to zero, and the subtitle in each panel reports $\kappa_{[0,100)}$. For all new construction, FAR is about 9% lower and dwelling units per acre about 14% lower immediately across the boundary on the more-stringent side. Among multifamily projects, FAR is about 13% lower and dwelling units per acre about 17% lower. All four differences are statistically distinguishable from zero at the 10% level or better. The similar results for FAR and dwelling units per acre indicate that the difference is present in both floor space and the number of homes placed on a parcel.

As a falsification test, I repeat the same comparison at artificial cutoffs 1,000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority. Online Appendix Figure D.1 shows little evidence of corresponding breaks at these artificial cutoffs. None of the eight placebo estimates is statistically distinguishable

from zero at the 10% level. The contrast with the true boundary makes it less likely that the main estimates reflect a general spatial pattern within wards. Online Appendix D presents the placebo figures along with continuity tests for fixed location characteristics, restrictions based on boundary geometry, and donut checks.

4.3 Evidence on Rents and Sale Prices

The permit and density results indicate that more stringent aldermen reduce development activity and the density of completed housing. The model in Section 3 indicates that this should feed into housing costs, the most directly policy-relevant outcome given Chicago’s current rent and price growth. For rents and sales, I estimate

$$\log(Y_{isp}) = \sum_{k \in \mathcal{K}, k \neq [-100, 0)} \kappa_k \mathbf{1}\{d_{is} \in k\} + X_i' \theta + \lambda_{t(i)} + \delta_{sp} + \varepsilon_{isp}, \quad (5)$$

where Y_{isp} is the real listed rent or real sale price for observation i near boundary segment s in calendar period p . As in the density analysis, $\mathcal{K}$ contains ten 100 ft distance bands and the band from -100 to 0 ft is omitted. The coefficient $\kappa_{[0, 100)}$ therefore compares observations in the two bands immediately across the boundary. The fixed effects compare observations near the same boundary segment in the same month for rents and the same quarter for sales. X_i includes unit or property characteristics and distances to nearby schools, parks, major roads, CTA stops, and Lake Michigan. The term $\lambda_{t(i)}$ denotes property-type fixed effects: building type reported in rental listings for rents and Cook County Assessor property class for sales. Standard errors are clustered by ward pair.

Figure 4 plots these distance-band estimates.

Listed rents are 1.8% higher on the more-stringent side relative to the less-stringent side, although the estimate is imprecise. Home sale prices are 3.0% higher on the more-stringent side relative to the less-stringent side. I treat the sales result as suggestive because transactions are less frequent near any particular boundary and their composition is harder

to hold fixed. Online Appendix E shows that the result disappears at artificial cutoffs inside either ward and is similar when I restrict the sample to straight boundaries or exclude observations immediately next to the boundary.

5 Discussion

5.1 Interpretation and Mechanisms

Taken together, the redistricting and boundary results show that individual politicians create meaningfully different regulatory environments within a single city. Within-census-block variation shows that permitting falls when a census block is reassigned to a more stringent alderman, while completed projects on the more-stringent side of ward boundaries are built at lower density. The price and rent comparisons are more tentative, but their direction is consistent with these differences in housing supply.

The central implication of these results is that the effect of political fragmentation on housing supply extends beyond formal zoning rules. Two otherwise similar projects governed by the same municipal code can face different timelines, negotiations, and prospects for approval because they are represented by different aldermen. Political fragmentation can therefore arise within a city, even without separate municipal governments or formally different regulatory codes.

These informal regulations can affect development along both extensive and intensive margins. Additional meetings, redesigns, and uncertain approval timelines reduce the expected return to development. Developers may respond to these additional demands, or the threat of them, by shrinking or abandoning projects before they enter the permitting data. The Fern Hill episode described in Section 2 illustrates these margins. The project that ultimately received approval was smaller than the original proposal, and approval followed four years of meetings and negotiation. The time-to-build framework in Section 3 captures the underlying economic force: delay lowers the value of development, reduces the scale of

projects that remain viable, and can cause marginal projects to disappear.

These findings complement Khan (2021), who shows that Chicago aldermen approve smaller increases in allowed density near ward boundaries, where their constituents capture fewer of the benefits from development. Informal political control can therefore affect not only what the zoning code allows, but also whether development proceeds and what ultimately gets built.

5.2 Informal Discretion Beyond Chicago

Chicago provides an especially clear setting because aldermanic privilege is unusually powerful, but the underlying institution is not unique to Chicago. Local council-member deference in land-use decisions has also been documented in Philadelphia, Los Angeles, and New York City (The Pew Charitable Trusts, 2015; Ryu, 2020; Anbinder, 2025; New York City Charter Revision Commission, 2025; Mitre-Becerril and MacDonald, 2024; Burnett and Kogan, 2014; Jacobson, 2023).

In many major U.S. cities, individual politicians can create distinct regulatory environments when projects require discretionary approval and other officials defer to the representative of the affected district. Under these conditions, the identity and preferences of the local politician can matter even when the written rules remain similar across districts. The broader lesson—that informal regulations can dampen housing supply and affect affordability—likely applies to other major cities as well.

5.3 Policy Implications

Local politicians may possess useful information about neighborhood conditions and provide residents with a clear channel for political accountability. But ward-level representatives also have little reason to internalize the citywide benefits of additional housing, and informal discretion allows them to impose delay and uncertainty after the formal rules have been established. The resulting regulatory environment may reflect the preferences of an individual

officeholder or a small group of constituents, without accounting for benefits elsewhere in the city.

While this paper cannot pin down the optimal balance between local input and centralized authority, it does point to predictability and transparency as important goals for reform. Expanding the scope of by-right approval would allow projects that comply with established rules to proceed without obtaining the support of an individual alderman. Published approval criteria, enforceable timelines, and meaningful appeal procedures could preserve local participation while limiting uncertainty. Moving some decisions to citywide bodies could also make it easier to account for housing benefits that extend beyond the ward in which a project is located.

Chicago’s lack of a city charter makes the allocation of authority particularly dependent on unwritten norms. A charter would not by itself eliminate aldermanic influence, but it could formally define approval authority, procedures, and avenues for appeal (McClintock, 2025). More generally, reforms aimed only at zoning classifications or formal voting rules may have limited effects if individual politicians retain informal control over permitting, project design, or administrative implementation.

These institutional choices matter as Chicago confronts rapidly rising rents (Davis, 2026). More broadly, the results show why evaluations of land-use regulation should look beyond the written zoning code. When individual political actors possess substantial discretionary authority, who administers the rules can matter as much as the rules themselves.

6 Conclusion

This paper studies how informal regulations and the discretion of individual political actors affect housing supply within a city. I focus on Chicago’s aldermanic privilege, the informal power of each of the city’s 50 aldermen to block or approve development in that member’s ward. This institution allows individual politicians to create different regulatory environ-

ments within a single municipal system.

Using permit processing times for projects where aldermen are most likely to matter, I construct aldermanic stringency scores that isolate the alderman-specific component of regulatory delay after accounting for permit, location, time, and ward characteristics. Using the 2015 ward redistricting to identify changes in assigned stringency, I find that reassignment toward greater stringency reduces the annual number of high-discretion permits recorded by the city by about 12% during 2015–2020. A boundary comparison shows that new construction is less dense on the more-stringent side of ward boundaries, with multifamily FAR about 13% lower and DUPAC about 17% lower immediately across the boundary. I also find a 3.0% increase in home prices and a suggestive 1.8% increase in listed rents, although the rent estimate is imprecise.

These results show that formal zoning rules do not fully describe the regulatory environment facing a developer. Similar projects governed by the same municipal and zoning codes can face different timelines and prospects for approval depending on the local politician. Effective land-use reforms must address not only the formal rules governing what can be built, but also the informal political institutions that determine whether and how development proceeds.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used OpenAI's Codex to assist with manuscript editing, primarily for clarity and concision, and with coding for data cleaning and analysis. The author reviewed and validated all code and edited all manuscript content as needed and takes full responsibility for the content of the published article.

References

- Achong, Ian.** 2025. “Shortened Version Of Old Town Canvas Approved By Alderman.” *Chicago YIMBY*.
- Anbinder, Jacob.** 2025. “New York NIMBYs Turn Against Democracy.” *The Atlantic*.
- Bayer, Patrick, Fernando Ferreira, and Robert McMillan.** 2007. “A Unified Framework for Measuring Preferences for Schools and Neighborhoods.” *Journal of Political Economy*, 115(4): 588–638.
- Bloom, Nicholas.** 2009. “The Impact of Uncertainty Shocks.” *Econometrica*, 77(3): 623–685.
- Bordeu, Olivia.** 2025. “Commuting Infrastructure in Fragmented Cities.” *Job Market Paper*.
- Burnett, Craig M., and Vladimir Kogan.** 2014. “Local Logrolling? Assessing the Impact of Legislative Districting in Los Angeles.” *Urban Affairs Review*, 50(5): 648–671.
- Chase, Brett.** 2023. “Chicago wrongly limited affordable housing with aldermanic prerogative, HUD says.” *Chicago Sun Times*.
- City of Chicago.** 2019. “Sixty-day Report on Implementation of Executive Order No. 2019-2.”
- Cunningham, Christopher R.** 2006. “House price uncertainty, timing of development, and vacant land prices: Evidence for real options in Seattle.” *Journal of Urban Economics*, 59(1): 1–31.
- Davis, Miranda.** 2026. “Chicago’s surging rents dent its cheap big city image.” *Bloomberg*.
- Duranton, Gilles, and Diego Puga.** 2023. “Urban Growth and Its Aggregate Implications.” *Econometrica*, 91(6): 2219–2259.

- Einhorn, R. L.** 1991. *Property Rules: Political Economy in Chicago, 1833-1872*. University of Chicago Press.
- Einstein, Katherine Levine, David M. Glick, and Maxwell Palmer.** 2019. *Neighborhood Defenders: Participatory Politics and America's Housing Crisis*. Cambridge University Press.
- Fischel, William A.** 2009. *The Homevoter Hypothesis: How Home Values Influence Local Government Taxation, School Finance, and Land-Use Policies*. Harvard University Press.
- Freemark, Yonah.** 2020. "Upzoning Chicago: Impacts of a Zoning Reform on Property Values and Housing Construction." *Urban Affairs Review*, 56(3): 758–789.
- Gallun, Alby.** 2019. "Ad hoc zoning, Chicago Style: How aldermen can rewrite the rules." *Crain's Chicago Business*.
- Giangreco, Leigh.** 2024. "Game of Zones." *Chicago Mag*.
- Glaeser, Edward L., Joseph Gyourko, and Raven Saks.** 2005. "Why Is Manhattan So Expensive? Regulation and the Rise in Housing Prices." *The Journal of Law and Economics*, 48(2): 331–369.
- Gyourko, Joseph, Jonathan S. Hartley, and Jacob Krimmel.** 2021. "The local residential land use regulatory environment across U.S. housing markets: Evidence from a new Wharton index." *Journal of Urban Economics*, 124: 103337.
- Hilber, Christian A.L., and Frédéric Robert-Nicoud.** 2013. "On the origins of land use regulations: Theory and evidence from US metro areas." *Journal of Urban Economics*, 75: 29–43.
- Hsieh, Chang-Tai, and Enrico Moretti.** 2019. "Housing Constraints and Spatial Misallocation." *American Economic Journal: Macroeconomics*, 11(2): 1–39.

- Jacobson, Emily.** 2023. “The Dysfunctional Metropolis: Reforming Los Angeles’s Land Use Planning and Entitlement.” California YIMBY Education Fund Report.
- Keefe, Alex.** 2013. “Pregnancy Tests? Pigeon Poo? What Chicago Aldermen Really Do.” *WBEZ Chicago*.
- Khan, Asad R.** 2021. “Decentralized Land-Use Regulation with Agglomeration Spillovers: Evidence from Aldermanic Privilege in Chicago.” *Job Market Paper*.
- Kline, Patrick, Evan K. Rose, and Christopher R. Walters.** 2022. “Systemic Discrimination Among Large U.S. Employers.” *The Quarterly Journal of Economics*, 137(4): 1963–2036.
- Kline, Patrick, Evan K. Rose, and Christopher R. Walters.** 2024. “A Discrimination Report Card.” *American Economic Review*, 114(8): 2472–2525.
- Kulka, Amrita, Aradhya Sood, and Nicholas Chiumenti.** 2026. “Under the (Neighbor)Hood: Understanding Interactions Among Zoning Regulations.” *The Review of Economics and Statistics*.
- Kydland, Finn E., and Edward C. Prescott.** 1982. “Time to Build and Aggregate Fluctuations.” *Econometrica*, 50(6): 1345.
- Lyons, Ronan C., and Éamonn Sweeney.** 2025. “Time to build: Rules-based planning and construction project duration in Dublin.” *Journal of Housing Economics*, 70: 102089.
- Mast, Evan.** 2024. “Warding off Development: Local Control, Housing Supply, and NIM-BYs.” *Review of Economics and Statistics*, 106(3): 671–680.
- Mayer, Christopher J, and C.Tsuriel Somerville.** 2000*a*. “Land use regulation and new construction.” *Regional Science and Urban Economics*, 30(6): 639–662.

- Mayer, Christopher J., and C.Tsuriel Somerville.** 2000*b*. “Residential Construction: Using the Urban Growth Model to Estimate Housing Supply.” *Journal of Urban Economics*, 48(1): 85–109.
- Mayor’s Press Office.** 2019. “Mayor Lightfoot Signs First Executive Order to Limit Aldermanic Prerogative.” Press Release.
- McClintock, Grant.** 2025. “What Is a City Charter, and How Could Having One Help Chicago?” *The Civic Federation*.
- Metropolitan Planning Council.** 2019. “Unfettered Aldermanic Prerogative.”
- Mitre-Becerril, David, and John M. MacDonald.** 2024. “Does urban development influence crime? Evidence from Philadelphia’s new zoning regulations.” *Journal of Urban Economics*, 142: 103667.
- Monarrez, Tomás, and David Schönholzer.** 2023. “Dividing Lines: Racial Segregation across Local Government Boundaries.” *Journal of Economic Literature*, 61(3): 863–887.
- New York City Charter Revision Commission.** 2025. “Charter Revision Commission Adopted Final Report.” City of New York.
- Oh, Hyunseung, Choongryul Yang, and Chamna Yoon.** 2024. “Land development and frictions to housing supply over the business cycle.” *Finance and Economics Discussion Series*, , (2024-010): 1–41.
- Ortalo-Magné, François, and Andrea Prat.** 2014. “On the Political Economy of Urban Growth: Homeownership versus Affordability.” *American Economic Journal: Microeconomics*, 6(1): 154–181.
- Paciorek, Andrew.** 2013. “Supply constraints and housing market dynamics.” *Journal of Urban Economics*, 77: 11–26.

- Ryu, David E.** 2020. “Limiting Unilateral Influence in Development Decisions.” Los Angeles City Council.
- Saiz, Albert.** 2010. “The Geographic Determinants of Housing Supply*.” *Quarterly Journal of Economics*, 125(3): 1253–1296.
- Schutz, Paris.** 2019. “Lightfoot, Aldermen Battle Over Aldermanic Prerogative.” *WTTW News*.
- Shriver Center on Poverty Law.** 2019. “A City Fragmented.”
- Soltas, Evan, and Jonathan Gruber.** 2026. “How Costly Is Permitting in Housing Development?” Working Paper.
- Soltas, Evan J.** 2024. “The Price of Inclusion: Evidence from Housing Developer Behavior.” *Review of Economics and Statistics*, 106(6): 1588–1606.
- Spielman, Fran.** 2019a. “Aldermen to be stripped of absolute power over licenses and permits, not zoning.” *Chicago Sun Times*.
- Spielman, Fran.** 2019b. “Lightfoot tells Burke to forget Council Wars 2: ‘That’s not going to happen’.” *Chicago Sun Times*.
- Thale, Christopher.** 2005. “Aldermanic Privilege.”
- The Pew Charitable Trusts.** 2015. “Philadelphia’s Councilmanic Prerogative.” Report.
- TRD Staff.** 2026. “Judge Tosses Lawsuit Challenging Old Town Apartment Tower Proposal.” *The Real Deal*.
- Turner, Matthew, Andrew Haughwout, and Wilbert Van Der Klaauw.** 2014. “Land Use Regulation and Welfare.” *Econometrica*, 82(4): 1341–1403.
- Zhang, Yue.** 2011. “Boundaries of Power: Politics of Urban Preservation in Two Chicago Neighborhoods.” *Urban Affairs Review*, 47(4): 511–540.

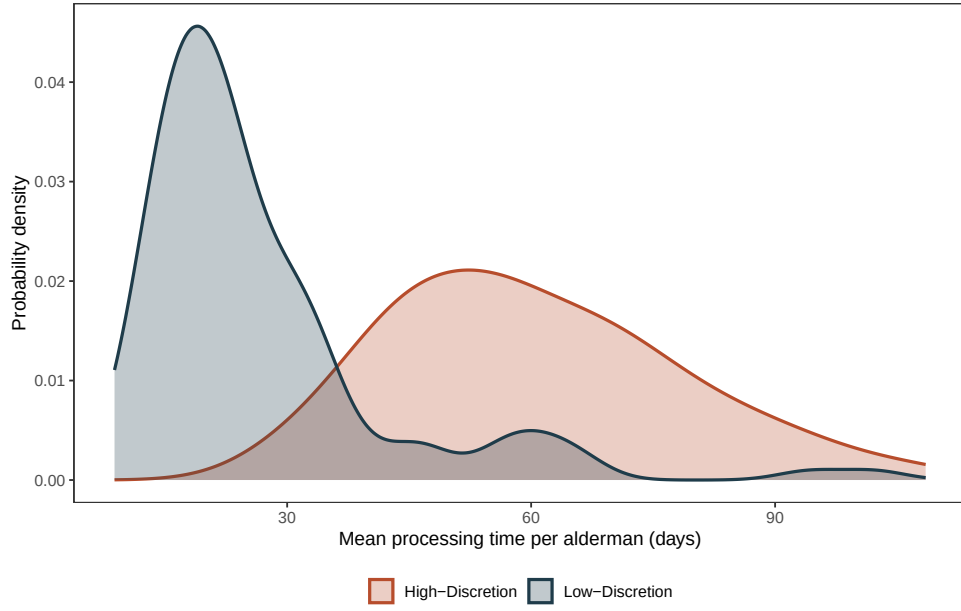


Figure 1: Permit Processing Times Across Aldermen

Notes: The curves show how mean processing times vary across aldermen, separately for high- and low-discretion permits, using records through December 2022. The high- and low-discretion permit groups are defined in Section 2.2. Processing time is days from application to issuance.

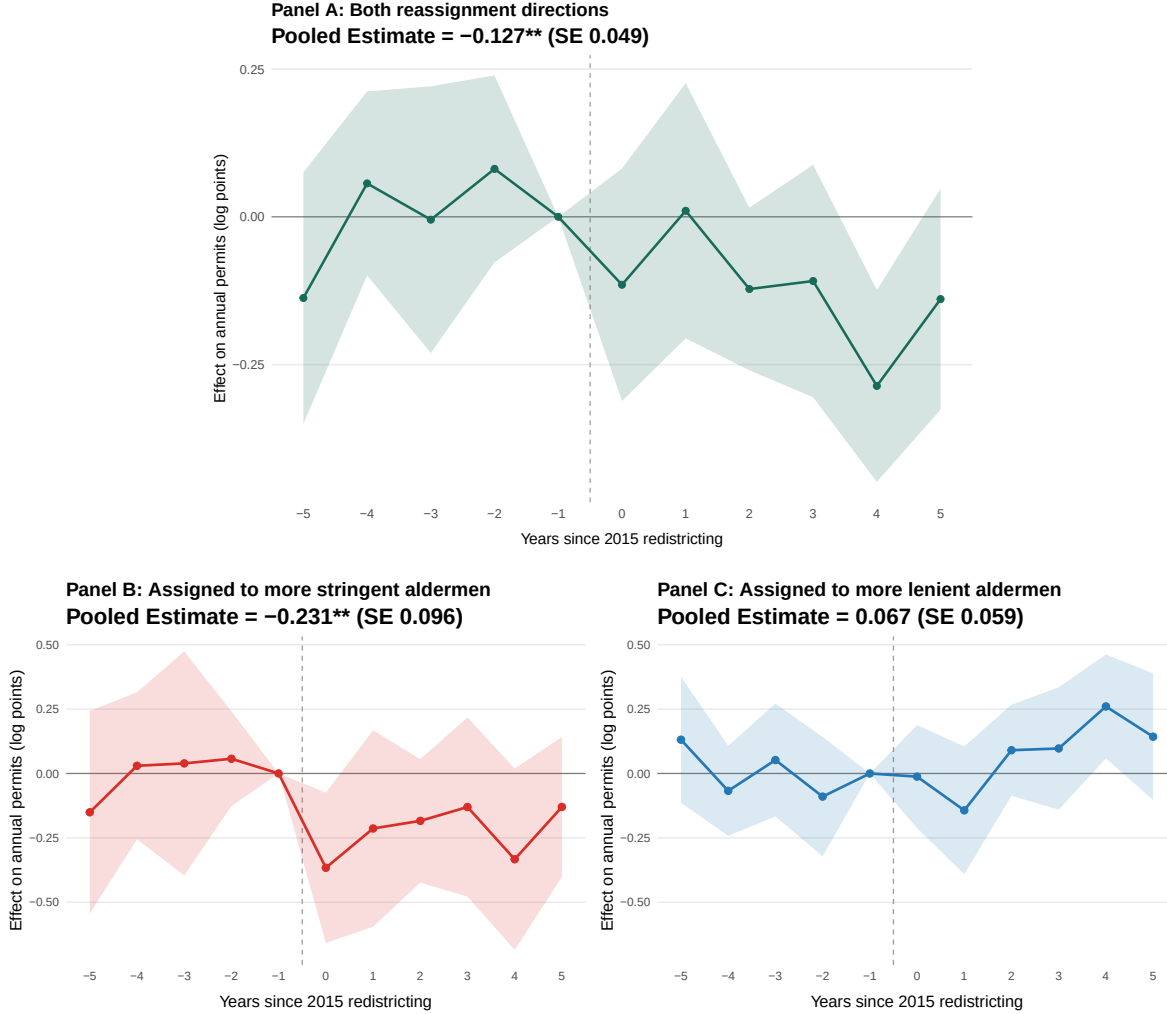


Figure 2: Permit Activity After the 2015 Ward Redistricting

Notes: The figure compares annual high-discretion permit activity on blocks reassigned to more- or less-stringent aldermen in 2015 with nearby blocks that stayed in their wards. The 2010–2020 sample includes blocks within 500 ft of ward boundaries; Online Appendix A.3 details sample construction. Panel A combines both reassignment directions using Equation (3); Panels B and C allow the effects of the two directions to differ. Points show effects in log points relative to 2014; subtitles report pooled 2015–2020 coefficients on the same scale. Panel A’s pooled coefficient of -0.127 corresponds to an 11.9% reduction in annual permits for assignment toward greater stringency. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. The test that the pre-2015 effects in Panel A are jointly zero gives $p = 0.400$. * $p \leq 0.10$, ** $p \leq 0.05$, *** $p \leq 0.01$.

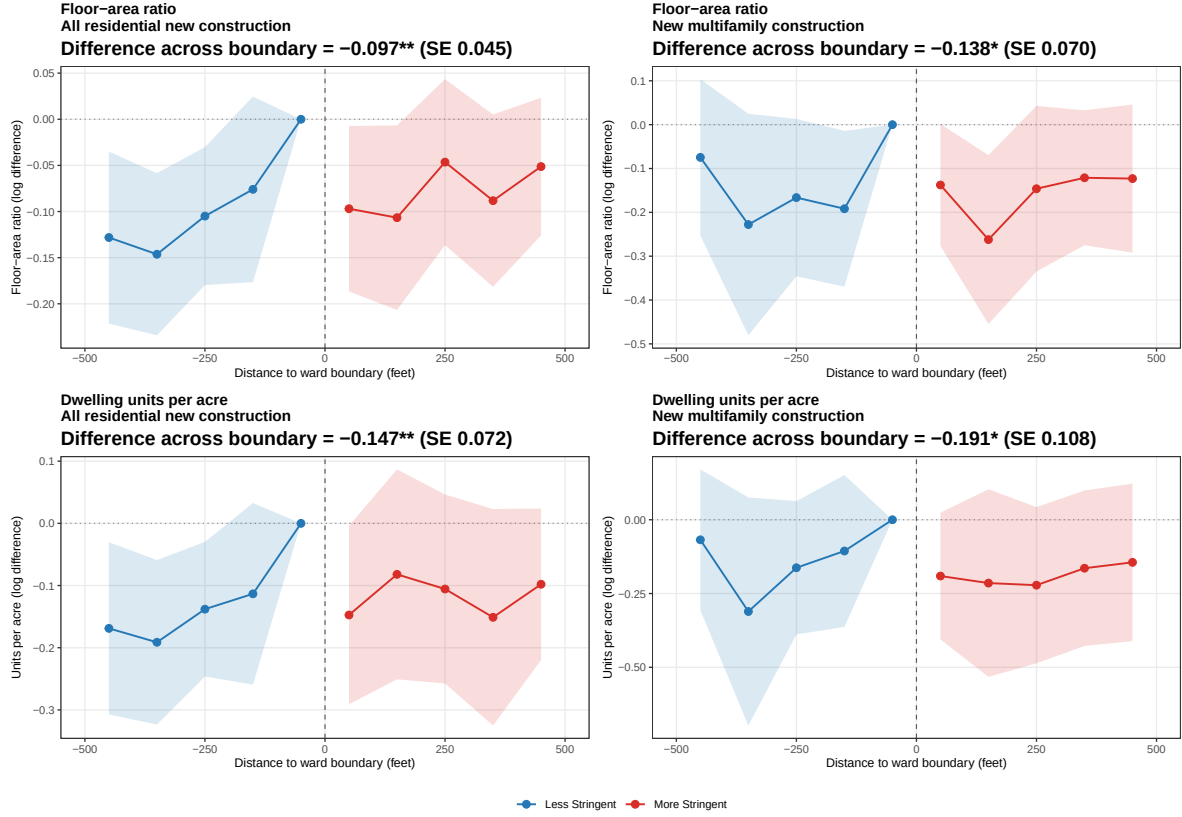


Figure 3: Development Density at Ward Boundaries

Notes: Left column: all residential new construction; right column: new multifamily construction. Top row: floor-area ratio; bottom row: dwelling units per acre, both in log differences. Samples include new construction within 500 ft of ward boundaries with both FAR and DUPAC available. Each point is the coefficient for a 100 ft distance band from Equation (4). The nearest band on the less-stringent side, from -100 to 0 ft, is omitted and normalized to zero. Subtitles report the coefficient for the nearest band on the more-stringent side, from 0 to 100 ft. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. Negative distance indicates the less-stringent side and positive distance the more-stringent side. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Listed rents and home sale prices at ward boundaries

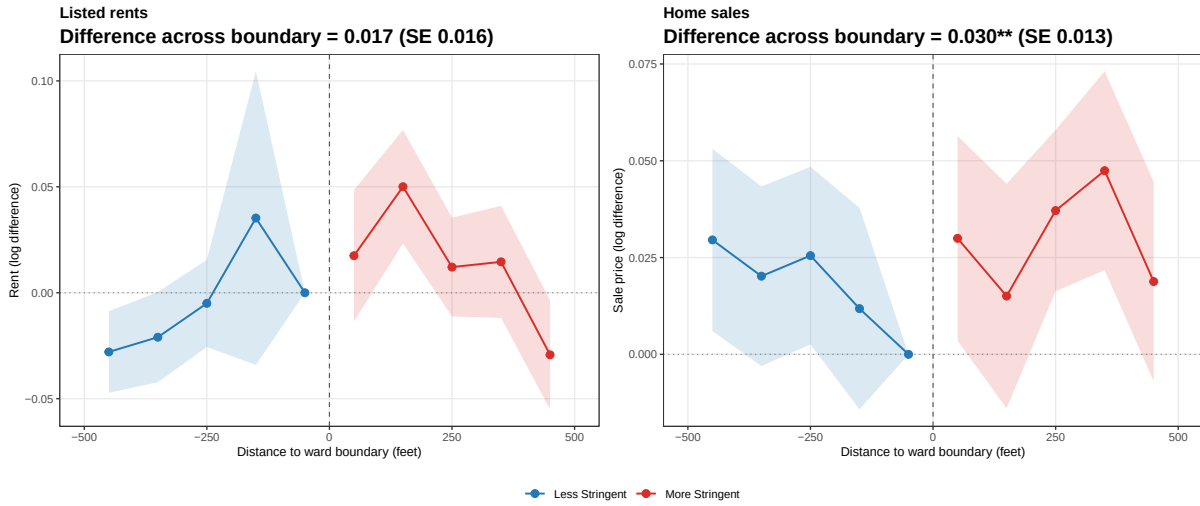


Figure 4: Listed Rents and Home Sale Prices at Ward Boundaries

Notes: The left panel uses monthly asking rents by floorplan from 2014–2022, with segment-by-month fixed effects. The right panel uses 58,468 sales of detached houses, townhouses, and apartment buildings with two to six units from 2006–2022, with segment-by-quarter fixed effects. Both panels control for housing characteristics and distances to nearby amenities. Rental controls include listing building-type fixed effects; sales controls include Cook County Assessor property-class fixed effects. Each point is the estimated log difference between a 100 ft distance band and the band immediately inside the less-stringent side, which is normalized to zero. Shaded areas are 95% confidence intervals. Subtitles report the comparison between the two bands adjacent to the boundary. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 1: New Residential Construction in Chicago, 2006–2022

	All New Construction	Multifamily
Average FAR	1.16	2.09
Average Units per Project	5.81	26.10
Average DUPAC	26.94	65.18
Median Distance to Boundary (ft)	915	826
Ward Pairs / Boundary Segments	139/1388	112/805
Number of Projects	13,635	2,582

Notes: The table summarizes projects with recorded building area, lot area, and dwelling units. FAR is building area divided by lot area. DUPAC is dwelling units per acre. Multifamily refers to buildings containing two or more dwelling units.

A Data and Sample Construction

This appendix provides additional detail on the data sources and sample rules described in Section 2.

A.1 Data Sources

The property data come from Cook County Assessor sales, assessment, parcel, and improvement files. RentHub provides rental listings from 2014–2022. Building permits and ward boundaries come from the City of Chicago. The amenity controls use school locations, Chicago Park District boundaries, major streets, CTA stops, and Lake Michigan geometry; CTA distances use the permanent station network in place during the listing month or on the sale date.

Demographic controls come from American Community Survey block groups and 2010 and 2020 census blocks. Because block groups do not follow ward boundaries, I assign their population and household counts to wards in proportion to the area that falls inside each ward. I use the resulting counts to calculate ward-level tenure, race, ethnicity, and education shares. The ward income measure is the household-weighted average of block-group median household income. Small overlaps in historical ward maps are divided proportionally so that no part of a block group is counted twice.

A.2 High-Discretion Permits

The permit records cover Department of Buildings permits since 2006. The high-discretion group contains the permit types defined in Section 2.2; low-discretion placebo permits are the remaining routine types, excluding signs. The event-study outcomes omit records marked cancelled, revoked, or suspended.

Processing time is the number of days between application and issuance. The processing-time regression excludes missing dates, negative durations, and same-day permits; same-day

permits remain in the lagged ward-month permit count. I assign each permit to the ward containing its coordinates under the boundaries in force on its application date.

A.3 Permit Event-Study Panel

The treatment panel compares each census block’s ward before and after the 2015 redistricting. I assign a block to the ward covering the largest share of its area. Let $a_o(b)$ denote the alderman representing block b under the old map and $a_d(b)$ the alderman under the new map. Blocks moved to a different ward are classified using the difference between these aldermen’s stringency scores:

$$\Delta S_b = S_{a_d(b)} - S_{a_o(b)}.$$

Positive values indicate assignment toward greater stringency, negative values indicate assignment toward greater leniency, and unchanged blocks form the comparison group.

For this design, stringency scores use permits filed from 2006 through 2014. Both origin and destination scores are assigned using the incumbents serving in 2014. The main sample retains blocks only when both incumbents remained in office when the new map took effect, so treatment matches the change the block experienced without using the outcome of the 2015 election. Section C.4 adds wards with electoral turnover and interprets the resulting estimate as assigned rather than necessarily realized stringency.

I count permit applications by block and year from 2010 through 2020. The sample includes blocks whose geographic center lies within 500 ft of a pre-redistricting ward boundary. For reassigned blocks, distance is measured to the boundary separating the old and new wards; for unchanged blocks, it is measured to the nearest boundary of the original ward. The estimation sample requires at least one high-discretion permit during event times $t = -5$ through $t = -1$.

A.4 New Construction

The construction sample combines Cook County residential improvement records with the separate Assessor records used for larger multifamily properties. I link records across parcel divisions, replacement PINs, and the two Assessor sources so that each project enters once. I then compare the resulting sample with every issued residential new-construction permit, adding completed projects absent from the Assessor files when units, building area, and lot area can be measured consistently.

Projects must have positive lot area, positive building area, at least one dwelling unit, and a construction year from 2006 through 2022. Assessor histories sometimes report different years or repeat whole-building characteristics across assessment cards. I use permit and completion dates to select the year that best matches completion, combine cards that clearly describe one building, and remove later PINs representing the same project. Where construction years or lot and building areas appear inconsistent in the Assessor files, I supplement these records with manual research using permits, property records, and other documented sources. The replication files record these decisions and their supporting evidence.

Locations first come from the Historical Parcel Universe in the construction year. When the original PIN is unavailable there, I use the current location of that PIN or a verified historical address. Each project retains one location, preventing predecessor and successor PINs from entering separately. Multifamily status primarily follows Assessor property class. When the reported unit count does not plausibly describe the whole building, permit descriptions and other property records determine the whole-building count, with these decisions also documented in the replication files.

Each project receives the broad zoning group in force on June 15 of its construction year. I combine official City zoning maps from 2012, 2014, and 2016 with passed map amendments dating to 2006, use annual parcel maps to follow sites whose PINs change, and retain planned developments as a separate zoning group. The reconstruction covers every project in the main and placebo boundary samples and reproduces the classifications observed in later official

zoning maps. Floor-area ratio (FAR) is total building area divided by lot area; dwelling units per acre (DUPAC) is the number of units divided by lot acreage. Projects are assigned to the ward map and alderman in office on June 15 of the reported construction year because the Assessor does not provide an exact completion date. I calculate ward assignments and distances in the Illinois State Plane coordinate system used for Chicago (EPSG:3435) and report distances in feet.

A.5 New-Construction Summary Statistics

Table 1 describes the citywide sample of 13,635 projects with both FAR and DUPAC available, including 2,582 multifamily projects.

A.6 Home Sales

The raw file contains 902,830 parcel-sale records in Chicago’s eight townships from 2006–2022. The transaction sample includes Assessor classes 202–210, 234, 278, and 295, which cover detached houses and townhouses, and class 211, which covers apartment buildings with two to six units. I exclude mixed-use class 212, condominiums, cooperatives, larger apartment buildings, residential land, and all other property classes.

Transactions must have a nominal price above \$10,000, not be explicitly identified as land-only, involve one parcel, use a warranty or trustee deed, and pass all three Assessor sale-quality flags: no duplicate sale of the same property within 365 days, no low-price flag, and no excluded-deed flag. When buyer and seller names are available, I exclude transfers with matching names. These transaction rules leave 238,705 sales with valid months and full 14-digit property identification numbers (PINs).

Some older records without an IDOR-refined date report the first day of the month even though the exact day is unavailable. I exclude these sales when either adjoining ward changes aldermen during that month, or when the ward map changes during the month, because the record does not identify the relevant alderman or ward map. This removes 1,289 transactions

citywide and leaves 237,416. I retain records with precise IDOR dates and unrefined dates that are not recorded on the first of the month.

I measure property characteristics using the Assessor record for the same property and tax year as the sale. I keep properties with one recorded building, positive building area, and an Assessor proration rate of one, which indicates that the record describes the entire property rather than a fractional share. These restrictions leave 233,744 sales. I then exclude 45 sales whose property records report more bedrooms than total rooms, leaving 233,699 sales. Missing apartment counts do not disqualify class-211 buildings because unit counts are not required by the price specification. When the sale and improvement files report different residential subclasses, I use the class recorded in the sale file for the property-class fixed effect. I express prices in 2022 dollars using the monthly Chicago CPI-U, with the mean 2022 index as the base, and do not trim or cap the price distribution.

I locate each transaction by matching its PIN to the parcel map for the year of sale; all 238,705 sales meeting the transaction-quality criteria receive an exact historical location. For retained transactions, ward, neighboring ward, and alderman assignments use the recorded sale date: transactions before May 18, 2015, use the 2003 ward map, and those on or after that date use the 2015 map. I calculate distance to the nearest boundary of the property's ward in the Illinois State Plane coordinate system. Distance is positive on the side represented by the more-stringent alderman and negative on the other side. I drop boundaries if either alderman lacks a stringency score or the two scores are identical.

For the boundary-price design, I split ward-pair boundaries into approximately 1,320-ft segments and assign each sale within 1,500 ft to the nearest segment on its ward-pair boundary. To compare similar homes in similar locations, the regression controls for log building area, log land area, log building age, log bedrooms, log bathrooms, a garage indicator, and distances to the nearest school, park, major road, CTA station open on the sale date, and Lake Michigan. Building age is sale year minus reported year built, and total bathrooms equal full bathrooms plus one-half of half bathrooms. The logged property characteristics

must be positive, so properties sold in their reported construction year are excluded because their measured age is zero. The preferred 500-ft sample contains 58,468 sales. Property-class fixed effects compare sales within the same Assessor class, while segment-by-sale-quarter fixed effects compare sales near the same short stretch of boundary in the same three-month period. Standard errors are clustered by ward pair.

A.7 RentHub Listings

The rental data record each occasion on which RentHub observed a listing. I standardize addresses and building types and treat blank or placeholder addresses as missing. Because RentHub does not report a consistent unit identifier, I group observations into properties using the standardized address and rounded coordinates, or coordinates alone when the address is missing. Within each property, I identify a floorplan by its bedrooms, bathrooms, square footage, and building type. I take the median asking rent when the same floorplan appears more than once on a date and then take the median across dates within each month. Before monthly aggregation, rent and square footage are trimmed at the 1st and 99th percentiles within year-by-bedroom cells.

The sample requires positive rent, valid coordinates, and assignment to a Chicago ward. Ward and alderman assignments use the listing's first observed date, with May 18 as the map transition date. The analysis excludes listings with coordinates that fail manual review, addresses that appear at inconsistent locations, and addressless listings concentrated at generic map points. It also excludes an address when replacing a listing's coordinate with that address's most frequently recorded location changes its ward assignment or moves its measured boundary distance by more than 100 ft. Regressions also require square footage, bathrooms, and amenity controls. Bedrooms enter categorically, so studios remain in the sample.

B Alderman Fixed Effects and Index Validation

This appendix provides additional detail on how the aldermanic stringency index is constructed and validated.

B.1 Construction of the Stringency Index

Aldermanic stringency measures whether comparable high-discretion permits take systematically longer under a particular alderman. It uses the permit sample described in Section A.2 through December 2022. Permits must take at least one day to enter the processing-time comparison, although same-day permits still count toward the ward’s recent permit volume.

Stage 1: Adjusting Processing Times. I first adjust processing times for observable reasons that otherwise could make permits slower in some wards. The regression controls for the permit’s distance to downtown and Lake Michigan, nearby CTA stations, ward demographics, recent ward permit volume, calendar month, permit type, and review type. Appendix A describes how block group demographics are allocated across ward boundaries to construct the ward controls.

Stage 2: Alderman Differences. I average the adjusted processing times by ward and month, then compare these averages across the aldermen serving each ward-month. Formally, I regress the ward-month averages on alderman indicators $\alpha_{a(w,m)}$ and the same recent permit-volume control. Ward-months are weighted by their permit counts. Here $a(w, m)$ denotes the alderman serving ward w in month m . The resulting fixed effect is the persistent alderman-specific difference in adjusted processing time.

Empirical Bayes Shrinkage. Some aldermen have fewer permits and therefore less precise estimates. I use empirical Bayes shrinkage to pull these noisier estimates toward the overall mean while leaving more precise estimates closer to their raw values: $\hat{\alpha}_a^{\text{EB}} = B_a \hat{\alpha}_a$,

where $B_a = \hat{\omega}^2 / (\hat{\omega}^2 + \hat{\sigma}_a^2)$. Here $\hat{\omega}^2$ is the estimated variance across aldermen and $\hat{\sigma}_a^2$ is the sampling variance of alderman a 's estimate.

Index Standardization. The stringency index, S_a , is standardized to have mean zero and unit standard deviation across aldermen. Higher values indicate more stringent aldermen.

Table B.1 reports the adjustment regression used in the first stage. The dependent variable is permit-level log processing time. These controls remove predictable differences in timing across locations and permit types before I compare aldermen. Panel B of Figure B.1 shows the distribution of the stringency scores.

B.2 Validation of the Stringency Index

The stringency index should measure differences across aldermen rather than permanent differences across wards. Panel C of Figure B.1 compares predecessor and successor aldermen within the same ward. If fixed ward characteristics drove the scores, aldermen serving the same ward at different times would receive similar values. The relationship is weak, which weighs against that explanation.

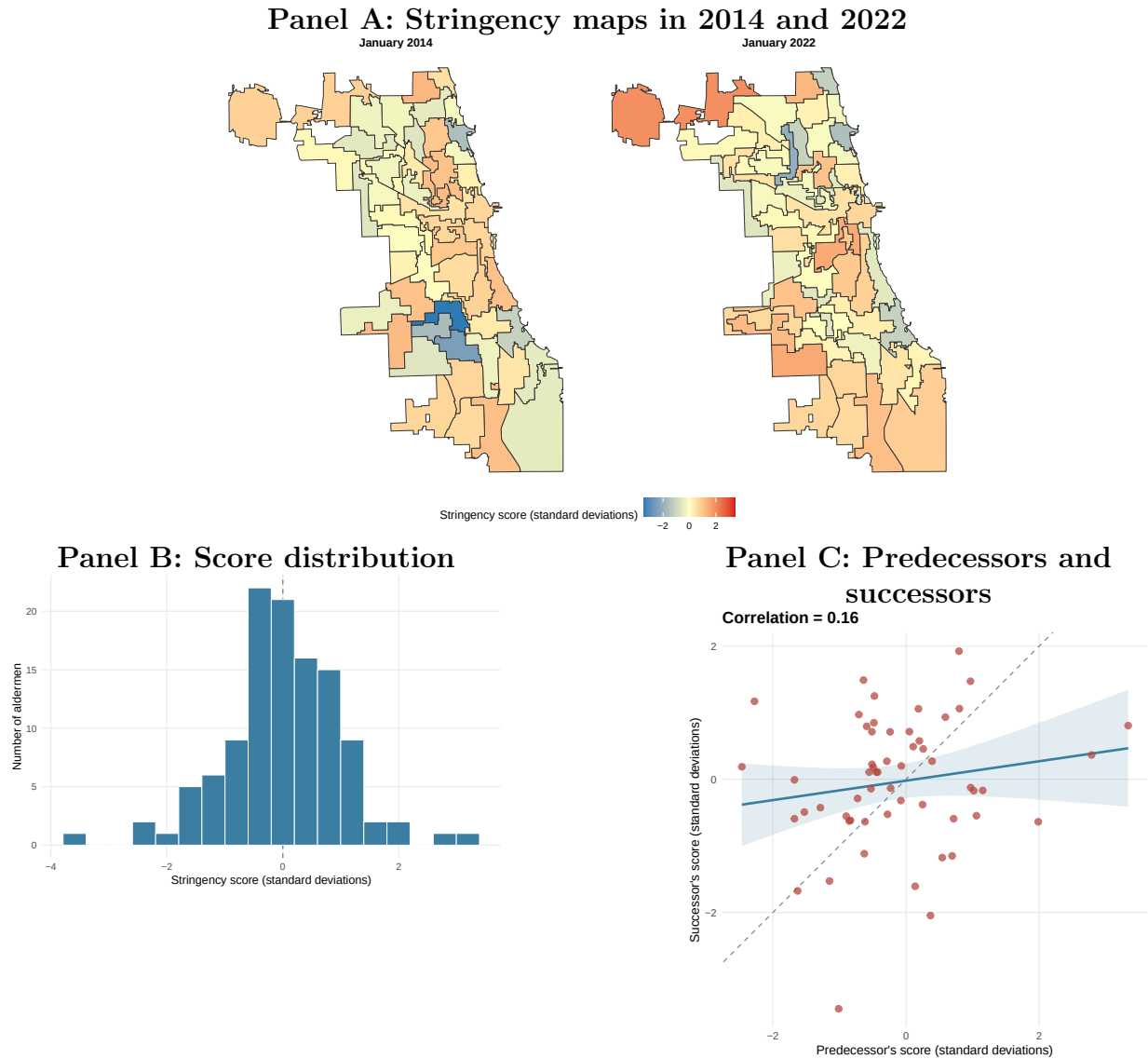


Figure B.1: Variation in Aldermanic Stringency

Notes: Stringency scores use permit processing times through December 2022 after adjusting for permit, location, ward, and calendar characteristics. Noisier alderman estimates are pulled toward the overall mean, and the resulting scores are standardized to mean zero and unit standard deviation. Panel A maps the scores of aldermen serving in January 2014 and January 2022 onto the corresponding ward boundaries; both maps use the same color scale, with warmer shades indicating more stringent aldermen. Panel B's dashed line marks the mean score of zero. Each point in Panel C pairs successive aldermen from the same ward. The dashed line indicates equal predecessor and successor scores; the solid line is a linear fit, with a shaded 95% confidence interval for the fitted mean.

Table B.1: Factors Associated with Permit Processing Times

	Log Processing Time (days) (1)
Median Household Income (\$10,000)	-0.015 (0.016)
Black Population Share	0.176 (0.278)
Hispanic Population Share	0.404 (0.272)
White Population Share	0.530 (0.352)
Homeownership Rate	0.261 (0.222)
Population (10,000)	-0.002 (0.020)
Distance to Downtown (km)	-0.017*** (0.006)
Distance to Lake (km)	-0.002 (0.005)
CTA Stations within 800 m	-0.009*** (0.003)
Permits in Previous Month	-0.002*** (0.0006)
Calendar Month Fixed Effects	✓
Permit Type Fixed Effects	✓
Review Type Fixed Effects	✓
N	120,505

Notes: Permit-level linear regression used to construct the stringency index, using permit records through December 2022. Covariates are median household income, racial shares, homeownership, population, distance to downtown, distance to the lake, nearby CTA stations, and the previous month's permit volume. Fixed effects are listed in the table. Regressions are unweighted. Standard errors are clustered by ward and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Additional Permit Evidence

This appendix reports additional permit evidence that complements the descriptive discussion in Section 2 and the event study in Section 4.1.

C.1 Balance Before Redistricting

Table C.1 compares blocks that changed wards in 2015 with nearby blocks that remained in the same ward. Before the new map took effect, the two groups were broadly similar in permit activity and neighborhood characteristics. The joint test across all listed characteristics has a p -value of 0.333.

C.2 Permit Event-Study Estimates

Table C.2 summarizes the average change in annual permit counts during the first six years under the new ward map. The combined model treats reassignment toward greater stringency and greater leniency as changes of equal size in opposite directions. A second model allows the two directions to have different effects. Low-discretion permits show a more muted response.

C.3 Allowing the Two Reassignment Directions to Differ

Panels B and C of Figure 2 allow moves toward greater stringency and greater leniency to have different effects. The two paths need not be mirror images, but their difference provides a comparable measure of how permit activity changes with stringency. Table C.2 shows that this comparison is nearly identical to the combined estimate.

Panel A of Figure C.1 shows the corresponding path for low-discretion permits. The response is more muted than for high-discretion permits.

C.4 Including Wards with Alderman Turnover

Panel B includes blocks from wards that elected new aldermen in 2015. The reassignment classification still uses the two aldermen serving in 2014. For blocks whose alderman subsequently changed, this measures the change implied by the map rather than the stringency the block necessarily experienced under the newly elected alderman. The annual number of high-discretion permits is 8.3% lower, a somewhat smaller estimate than in the main sample.

C.5 Permit Summary Statistics and Volume

Table C.3 compares the high- and low-discretion permit groups and reports their ward-month processing-time and permit-volume correlations. High-discretion permits are much slower than lower-discretion permits: the mean processing time is 57.3 days versus 26.2 days, and the median is 30 days versus 6 days. They also display more cross-alderman variation in processing times. Ward-months with longer high-discretion processing times have fewer high-discretion permits. Low-discretion permits show a weak positive relationship instead.

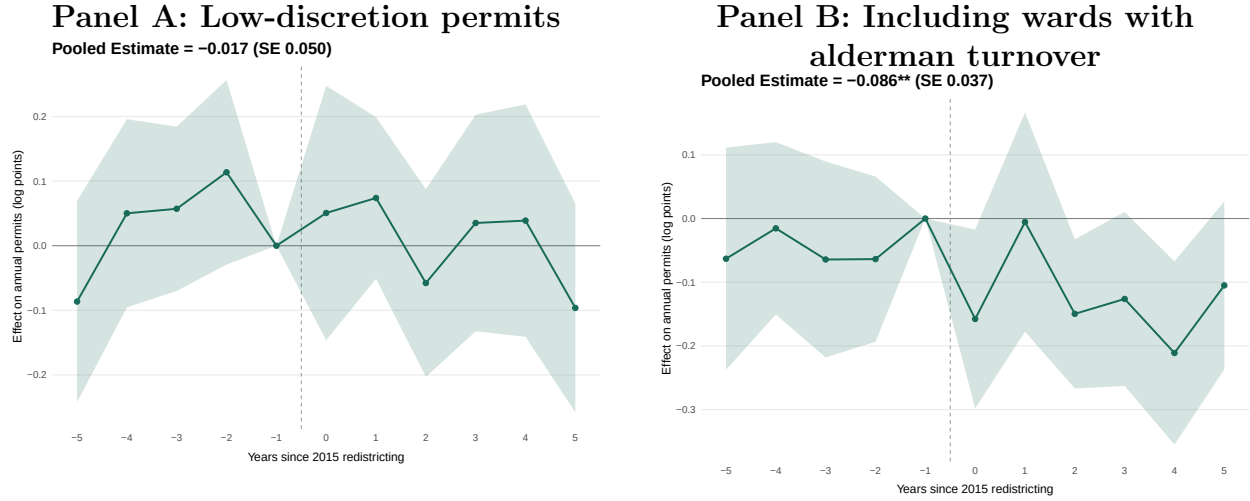


Figure C.1: Additional Permit Event-Study Checks

Notes: Panel A uses low-discretion permits excluding signs. Panel B uses high-discretion permits and includes wards with alderman turnover; assignment still uses the 2014 incumbents. Both panels combine the two reassignment directions using Equation (3); negative coefficients indicate fewer permits under assignment toward greater stringency. Points show effects in log points relative to 2014; subtitles report pooled 2015–2020 coefficients on the same scale. The pooled effects correspond to reductions of 1.7% and 8.3% in annual permits, respectively. Tests that the pre-2015 effects are jointly zero give $p = 0.075$ in Panel A and $p = 0.666$ in Panel B. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p \leq 0.10$, ** $p \leq 0.05$, *** $p \leq 0.01$.

Table C.1: Pre-Redistricting Balance Between Reassigned and Unchanged Blocks

Variable	Unchanged	Reassigned	Difference	SE	<i>p</i> -value	<i>N</i>
<i>Permit history</i>						
High-discretion permits, 2010–2014	2.668	2.649	0.086	0.081	0.291	2,340
Low-discretion permits, 2010–2014	4.278	4.377	0.146	0.412	0.725	2,340
No low-discretion permit, 2010–2014	0.114	0.127	0.008	0.023	0.713	2,340
New-construction permits, 2010–2014	0.381	0.349	0.013	0.044	0.766	2,340
No new-construction permit, 2010–2014	0.773	0.798	-0.008	0.013	0.525	2,340
<i>Neighborhood characteristics</i>						
Median household income	58,210	53,228	-1,352	1,354	0.324	2,324
Homeownership rate	0.404	0.407	0.000	0.011	0.998	2,340
Bachelor’s degree or higher share	0.441	0.399	-0.000	0.009	0.970	2,340
Black population share	0.301	0.272	0.003	0.006	0.667	2,340
Hispanic population share	0.226	0.276	-0.011	0.009	0.230	2,340
Median gross rent	1,068	1,022	-19	16	0.246	2,285
Median home value	309,400	283,787	-8,561	11,019	0.441	2,288
Average household size	2.550	2.694	-0.004	0.052	0.932	2,340
Median age	33.8	33.5	0.1	0.4	0.798	2,340
Population per square kilometer	8,350	8,187	-100	378	0.791	2,340

Notes: Reassigned blocks moved to a different ward under the 2015 map; unchanged blocks remained in their original ward. The sample requires the 2014 origin- and destination-ward incumbents to remain in office when the new map took effect. It also requires at least one high-discretion permit during 2010–2014. The sample contains 687 reassigned and 1,653 unchanged blocks within 500ft of a ward boundary in 44 ward pairs containing both groups. Permits are grouped by application year. Neighborhood characteristics are 2014 ACS five-year block-group estimates. Difference compares reassigned and unchanged blocks within the same ward pair; standard errors are clustered by ward pair. Means are unadjusted block-level means. The joint-test *p*-value for all listed covariates is 0.333 (complete-case $N = 2,249$).

Table C.2: Permit Effects with Combined and Separate Reassignment Directions

	High-Discretion	Low-Discretion
Combined reassignment effect	-0.127** (0.049)	-0.017 (0.050)
<i>Directions estimated separately</i>		
Assigned to more stringent aldermen	-0.231** (0.096)	0.037 (0.074)
Assigned to more lenient aldermen	0.067 (0.059)	0.043 (0.065)
One-half difference between directions	-0.149*** (0.052)	-0.003 (0.048)
Equal-and-opposite test p -value	0.180	0.432
Block fixed effects	Yes	Yes
Ward-pair $\times$ year fixed effects	Yes	Yes
Positive pre-period permit activity required	Yes	Yes
N	36,830	35,541

Notes: Poisson regressions use census-block-year observations from 2010–2020 within 500 ft of a ward boundary. Stringency scores are estimated using permits filed from 2006 through 2014. The sample retains blocks whose 2014 origin- and destination-ward incumbents remained in office when the new map took effect. It also requires at least one high-discretion permit during event times $t = -5$ through $t = -1$. Blocks are classified as assigned toward greater stringency, assigned toward greater leniency, or unchanged under the new map. The combined model imposes equal-and-opposite effects for the two reassignment directions, with unchanged blocks as the reference group; a negative coefficient means that greater stringency reduces permits. The separate model includes one indicator for each reassignment direction. Its half-difference row puts the two-direction estimates on the same scale as the combined estimate. The final test asks whether the separately estimated effects are equal in size and opposite in sign. Each model includes block fixed effects and ward-pair $\times$ year fixed effects. The estimator omits blocks with no permits of the relevant type in any sample year and ward-pair-year groups with no such permits. Standard errors, in parentheses, are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Processing-time summary statistics

Statistic	High-Discretion	Low-Discretion
Total permits in sample	121,405	124,124
Mean processing time (days)	57.3	26.2
Median processing time (days)	30.0	6.0
Alderman-level mean processing time (days)	60.2	26.6
Alderman-level IQR of mean processing time (days)	24.5	13.5
Number of unique wards	50	50
Number of unique aldermen	114	114

Panel B: Ward-month processing time and permit volume

Permit group	Correlation with ward-month permit volume
High-Discretion	-0.123
Low-Discretion	0.106
All	-0.021

Table C.3: Summary Statistics for High- and Low-Discretion Permits

Notes: The sample uses permits issued at least one day after application, with applications begun through December 2022. The permit groups are defined in Section 2.2. The alderman-level interquartile range is computed across alderman-specific mean processing times. Panel B reports Pearson correlations. Each observation represents one ward in one month.

D Additional Density Boundary Checks

This appendix asks whether the density result could be an artifact of how I measure aldermanic stringency or where ward boundaries happen to lie. It also tests for jumps away from true boundaries and checks whether projects immediately next to a boundary drive the result. Unless otherwise noted, each estimate compares the two 100 ft bands immediately across the ward boundary using Equation (4). All checks use projects with both FAR and DUPAC available.

D.1 Stringency-Score Checks

The first check asks whether a project's own permits influence the score used to classify its boundary. For each project, I recalculate the scores for the two aldermen after removing permits matched to that project. The estimates are effectively unchanged. The second check drops boundaries where the two aldermen receive similar scores and the more-stringent side is therefore harder to distinguish. The estimates of FAR and units per acre remain negative in both construction samples when the required score difference is either 0.25 or 0.50 standard deviations, although the multifamily estimates become less precise as the sample shrinks.

D.2 Placebo Cutoffs

Figure D.1 moves the cutoff 1,000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority. None of the eight placebo estimates is statistically distinguishable from zero at the 10% level.

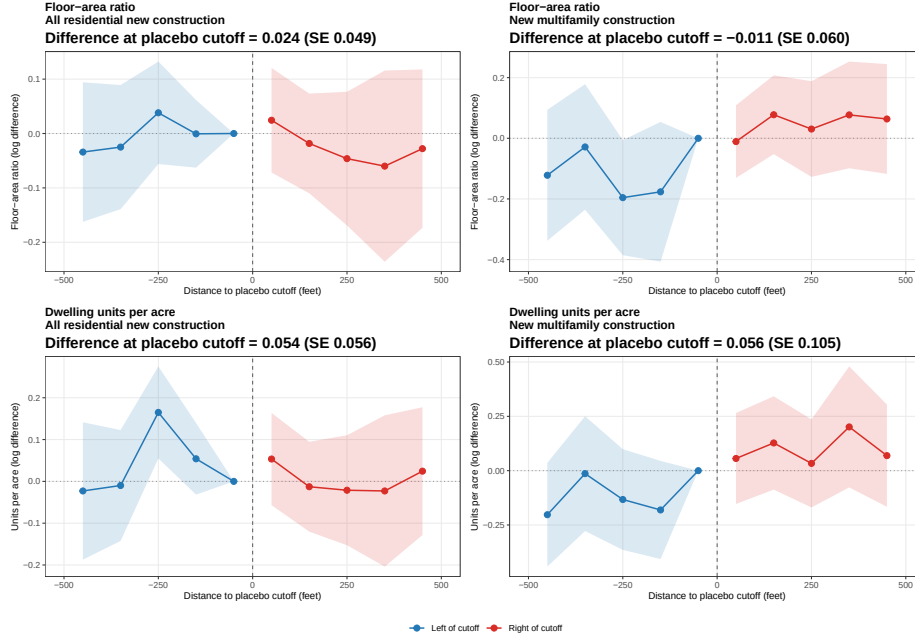
D.3 Continuity in Location Characteristics

Table D.2 tests whether fixed features of a project's location change at the ward boundary. None of the differences in distance to downtown, the nearest school, the nearest park, or Lake Michigan is statistically distinguishable from zero in either sample.

D.4 Boundary Geometry

Some ward boundaries follow expressways, waterways, arterial roads, or other physical features. Table D.3 therefore repeats the analysis after restricting the sample based on the geometry of the assigned boundary segment. Removing segments with substantial expressway or water overlap leaves the estimates almost unchanged. A broader physical-feature and arterial-road restriction attenuates the multifamily estimates, while the all-construction estimates remain negative. Restricting the sample to locally straight boundaries also produces negative estimates for FAR and units per acre in both construction samples. These estimates are generally smaller and less precise in the reduced sample, but they do not suggest that boundary curvature alone creates the main pattern.

Panel A: 1,000 ft inside the less-stringent side



Panel B: 1,000 ft inside the more-stringent side

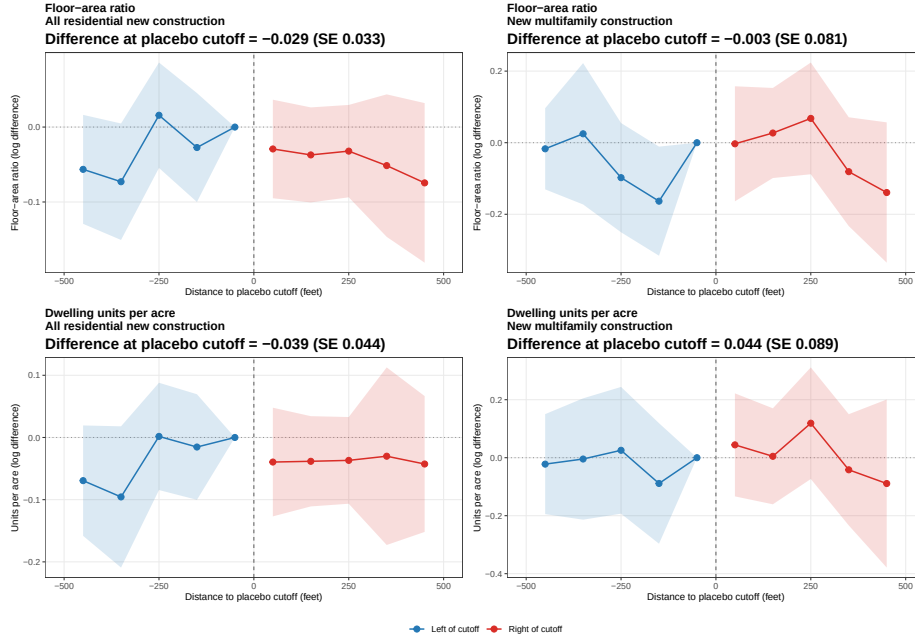
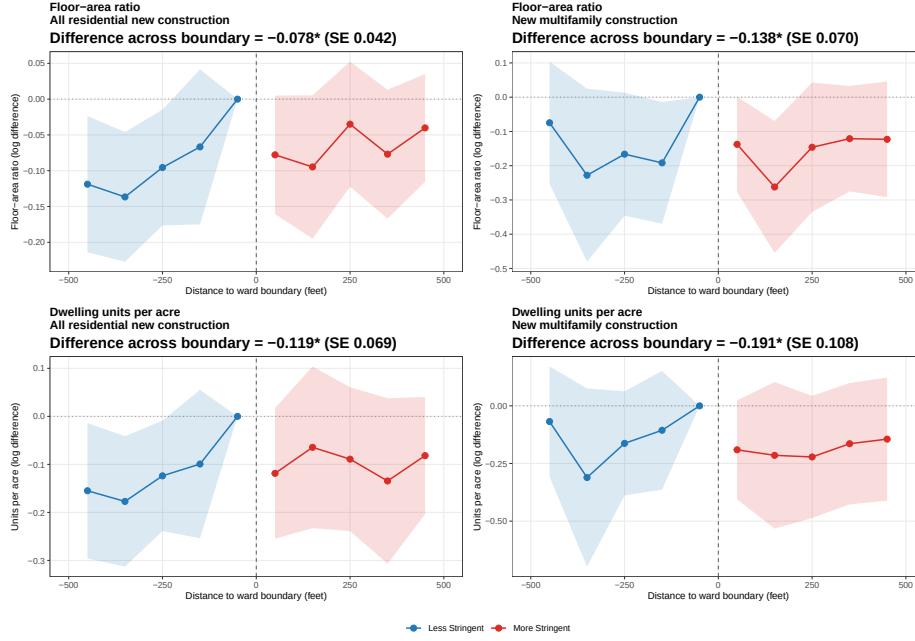


Figure D.1: Density Estimates at Placebo Cutoffs

Notes: Each artificial cutoff lies 1,000 ft inside the indicated side of the true ward boundary. The model uses a 500 ft window around the artificial cutoff and the same 100 ft distance bands, controls, and fixed effects as Equation (4). The 100 ft band just below the artificial cutoff is the reference group; subtitles report the difference for the 100 ft band just above it. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair.

Panel A: Excluding projects within 25 ft



Panel B: Excluding projects within 50 ft

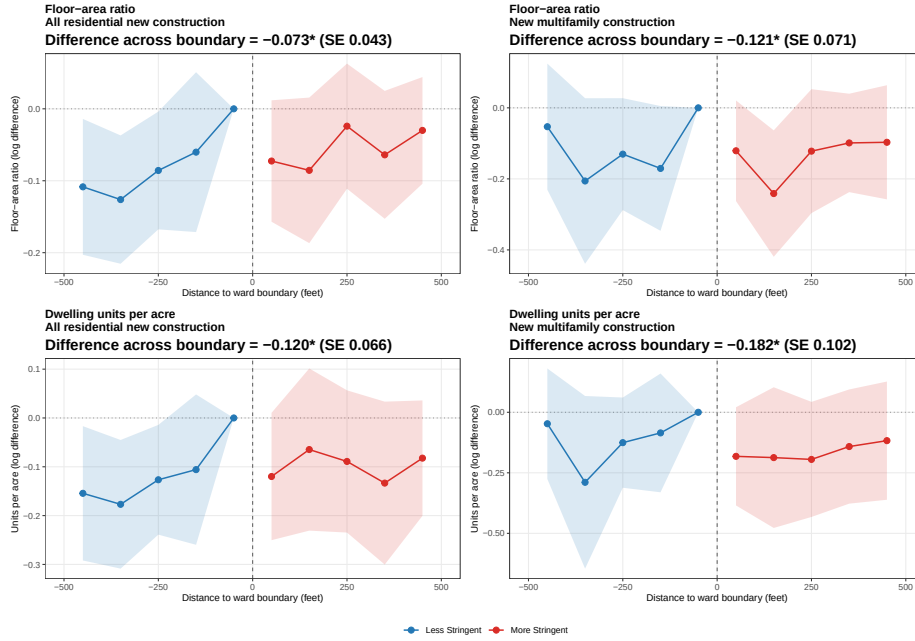


Figure D.2: Density Estimates Excluding Projects Nearest the Boundary

Notes: This check asks whether projects immediately adjacent to a boundary drive the main result. Each panel excludes projects within the indicated distance and otherwise uses the 500 ft sample and Equation (4). After each exclusion, the remaining part of the -100 to 0 ft band on the less-stringent side is the reference group. In each panel, the estimated difference across the boundary remains negative for FAR and units per acre in both construction samples. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.1: Density Estimates Under Alternative Measures of Boundary Stringency

	All Construction		Multifamily	
	Log floor-area ratio	Log units per acre	Log floor-area ratio	Log units per acre
<i>Panel A: Removing permits linked to each project</i>				
Full-sample score	-.097** (.045)	-.147** (.072)	-.138* (.070)	-.191* (.108)
N	4,023	4,023	863	863
Ward pairs/segments	122/873	122/873	88/425	88/425
Score excluding the project's permits	-.097** (.045)	-.147** (.072)	-.138* (.070)	-.191* (.108)
N	4,023	4,023	863	863
Ward pairs/segments	122/873	122/873	88/425	88/425
<i>Panel B: Excluding boundaries with similar scores</i>				
Score difference at least 0.25 SD	-.118** (.059)	-.171** (.078)	-.066 (.084)	-.070 (.109)
N	2,964	2,964	710	710
Ward pairs/segments	118/771	118/771	80/357	80/357
Score difference at least 0.50 SD	-.122* (.064)	-.172** (.085)	-.153 (.105)	-.141 (.124)
N	2,313	2,313	530	530
Ward pairs/segments	100/582	100/582	64/273	64/273

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The second row removes permits matched to each construction project before recalculating the alderman scores used for that project. The score-difference rows retain boundaries where the absolute difference between the two alderman scores is at least the stated number of standard deviations. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.2: Continuity in Location Characteristics at Ward Boundaries

	All Construction	Multifamily
Distance to downtown (miles)	0.001 (0.009)	-0.009 (0.012)
Distance to nearest school (miles)	0.007 (0.005)	-0.002 (0.009)
Distance to nearest park (miles)	-0.006 (0.006)	-0.011 (0.010)
Distance to Lake Michigan (miles)	0.007 (0.007)	0.002 (0.011)
Observations	4,023	863

Notes: Each row uses the listed location characteristic as the dependent variable. The reported coefficient compares the 0–100 ft band on the more-stringent side with the –100 to 0 ft band on the less-stringent side. Regressions include 100 ft distance-band indicators, boundary-segment fixed effects, and construction-year fixed effects. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.3: Density Estimates Along Different Types of Ward Boundaries

	All Construction		Multifamily	
	Log floor-area ratio	Log units per acre	Log floor-area ratio	Log units per acre
Main sample	-0.097** (0.045)	-0.147** (0.072)	-0.138* (0.070)	-0.191* (0.108)
Observations	4,023	4,023	863	863
Limited expressway or water overlap	-0.096** (0.045)	-0.146** (0.072)	-0.139* (0.070)	-0.193* (0.109)
Observations	3,769	3,769	840	840
Limited physical-feature or arterial overlap	-0.082* (0.045)	-0.128* (0.068)	-0.135* (0.078)	-0.152 (0.096)
Observations	3,181	3,181	690	690
Straight boundary segment	-0.081* (0.046)	-0.125* (0.065)	-0.109 (0.081)	-0.161 (0.121)
Observations	3,140	3,140	625	625

Notes: Entries compare the two 100 ft bands immediately across the boundary using Equation (4). The expressway-or-water restriction removes segments for which either feature overlaps at least half of the segment. The broader restriction removes segments with at least 50% park, water, or cemetery overlap, 40% expressway overlap, or 75% arterial-road overlap. A boundary is classified as locally straight when a 328 ft straight line centered on the project's nearest boundary point stays within 49 ft of the actual boundary at both ends. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Additional Price Boundary Checks

Prices at artificial boundaries

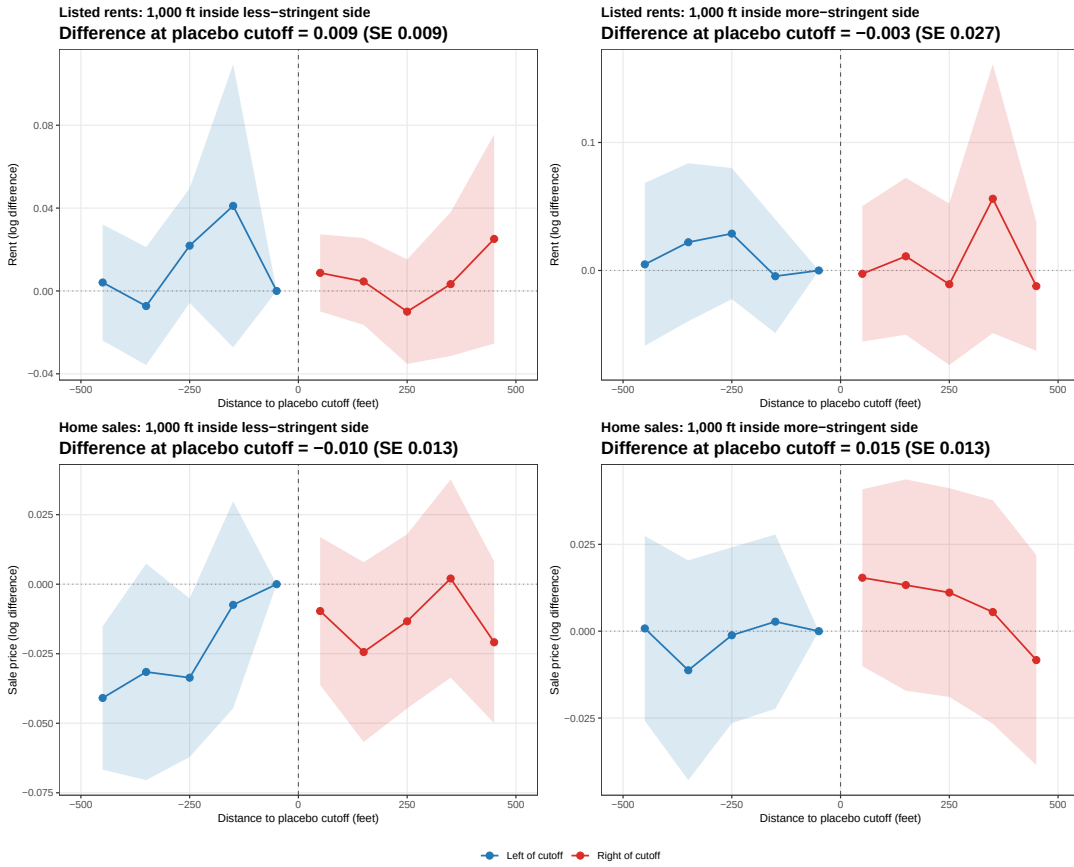
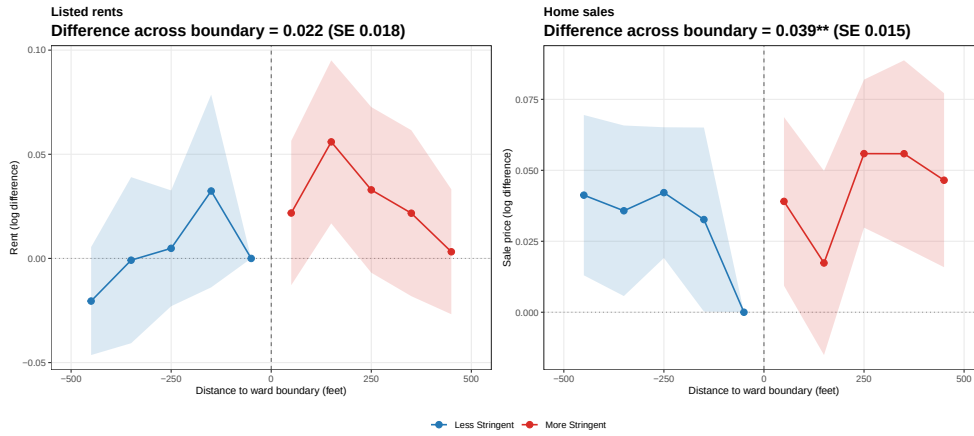


Figure E.1: Placebo Cutoffs for Listed Rents and Home Sale Prices

Notes: Neither rents nor sale prices change significantly across an artificial cutoff. The left column places an artificial cutoff 1,000 ft inside the less-stringent ward; the right column places it 1,000 ft inside the more-stringent ward. The top row reports listed rents and the bottom row reports home sale prices. Each model uses a 500 ft window around the artificial cutoff and otherwise follows Equation (5). Each point is the estimated log difference between a 100 ft distance band and the band immediately below the artificial cutoff. Subtitles compare the bands immediately above and below the cutoff. Shaded areas are 95% confidence intervals. Standard errors are clustered by ward pair.

Panel A: Locally straight boundaries



Panel B: Excluding observations nearest the boundary

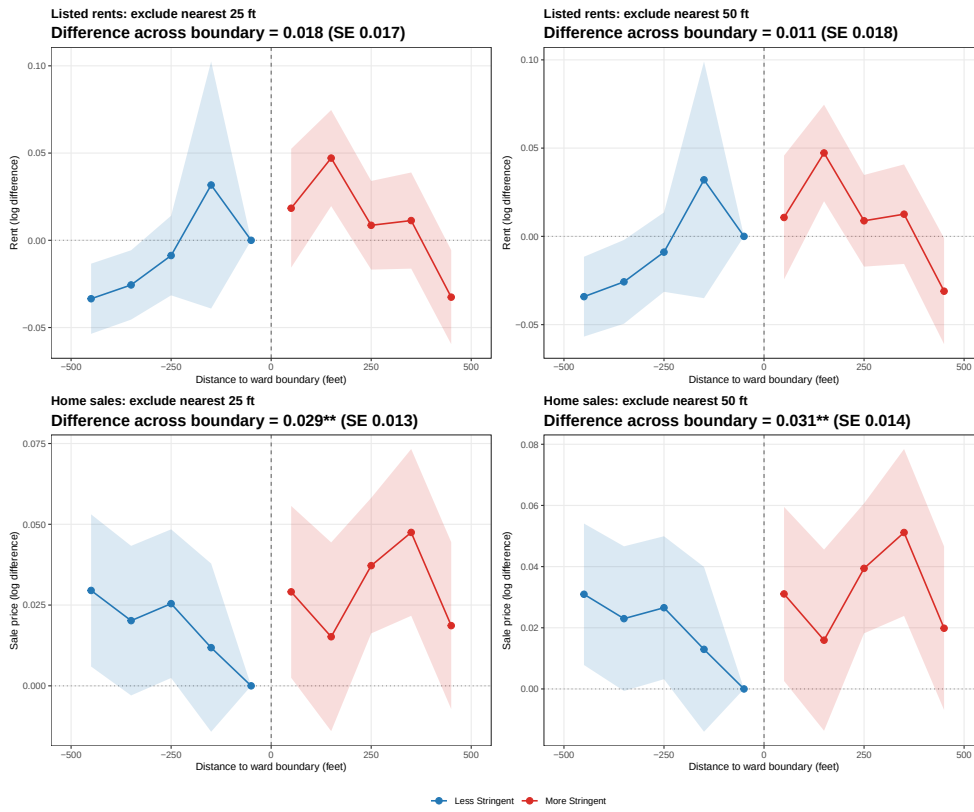


Figure E.2: Additional Price-Boundary Robustness Checks

Notes: Panel A retains locally straight boundaries: a 328 ft line centered on the nearest boundary point stays within 49 ft of the boundary at both ends. Panel B excludes observations within 25 ft (left) or 50 ft (right); the top row reports listed rents and the bottom row home sale prices. All models otherwise follow Equation (5). Points report log differences from the nearest 100 ft band on the less-stringent side, normalized to zero; subtitles compare the bands across the boundary. Panel B uses the remaining observations in each band after exclusion. Shaded areas are 95% confidence intervals, and standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.